



MTH204 : Symmetry (2024-25)

Tutorial 08 (April 21, 2025)

1. Define symmetry for a subset of $\mathbb{R}^2$ and $\mathbb{R}^3$. Give examples.
2. Draw a shape in $\mathbb{R}^2$ with exactly 3 rotational symmetries but no reflection symmetries.
3. Draw a shape in $\mathbb{R}^2$ with exactly 1 reflection symmetry.
4. Show that the composition of two reflections about parallel lines is a translation.
5. List all rotational symmetries of a tetrahedron.
6. Find a symmetry of cube which is not a rotational symmetry.
7. Write the 3×3 matrix for 120° rotation about the axis $(1, 1, 1)$.
8. Find the stabilizer of a vertex under rotational symmetries of a cube.
9. Show that A_4 can be generated by its elements of order 3.
10. Draw the Cayley graph of S_3 with generators $\{(12), (123)\}$.
11. A non identity diagonal matrix M represents a rotation in three dimensions. Then show that the sum of diagonal elements of M is negative.
12. Is the group of rotational symmetries of a tetrahedron a subgroup of the group of rotational symmetries of a dodecahedron? Justify your answer.
13. Consider the unit quaternion $q = \frac{1}{2}(1 + i + j + k)$. Explicitly express in terms of axis and rotation angle, the rotation in $\mathbb{R}^3$ represented by q .
14. Consider the line $\mathcal{L}$ represented by the equation $y = x$ in $\mathbb{R}^2$. Let $f_{\mathcal{L}}$ denote the reflection about $\mathcal{L}$, f_X denote the reflection about X -axis and f_Y denote the reflection about Y -axis.
 - (a) Let $P = (a, b) \in \mathbb{R}^2$. If $f_{\mathcal{L}}(P) = (a', b')$ then determine a' and b' in terms of a and b .
 - (b) Let G denote the group generated by $\{f_X, f_Y, f_{\mathcal{L}}\}$. Determine the order of G . Is G abelian?
 - (c) For the action of the group G as in (b) above on $\mathbb{R}^2$ determined by how $f_X, f_Y, f_{\mathcal{L}}$ move points of $\mathbb{R}^2$, find all possible orbit sizes.
15. Give examples of three non-identity 2×2 matrices A, B and C such that $A^2 = B^2 = C^2 = I_{2 \times 2}$ and all the three matrices commute with each other. (Hint: Klein's 4-group is a subgroup of $SO(3)$. You may also solve it without this hint.)

16. Vertices of squares of identical size are coloured using a color palette of red, blue and green colors to form various coloured configurations. Two coloured configuration of these squares are said to be from the same coloured toy if one of them may be rotated or flipped suitably so that the two become indistinguishable. How many distinct families of coloured toys can be made this way?
17. (a) Find a unit quaternion that determines the rotation in $\mathbb{R}^3$ about the axis $u = (1, 1, 1)$ by the angle $\theta = 180^\circ$.
- (b) Denote the rotation as in (a) by $\mathcal{R}$. If $P = (1, 2, 4)$ then determine the coordinates of $\mathcal{R}(P)$.